

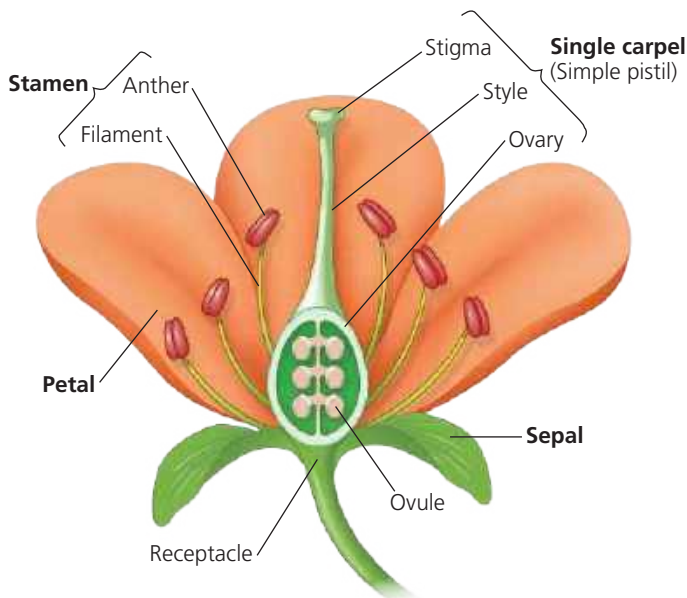
In Chapters 29 and 30, we approached plant reproduction from an evolutionary perspective, tracing the descent of plants from algal ancestors. Because angiosperms are the most important group of plants in most terrestrial ecosystems, we'll examine their reproductive biology, both sexual and asexual, in detail in this chapter. The importance of angiosperm reproduction, however, is not limited to natural ecosystems: the cultivation of angiosperms forms the basis for much of agriculture. For over 10,000 years, plant breeders have genetically manipulated the traits of a few hundred wild angiosperm species by artificial selection, transforming them into the crops we grow today. In recent decades, genetic engineering has dramatically increased the variety of ways and the speed with which we can modify plants, but many controversies have followed in the wake of this new biotechnology.

CONCEPT 38.1

Flowers, double fertilization, and fruits are key features of the angiosperm life cycle

The life cycles of all plants are characterized by an alternation of generations, in which multicellular haploid (n) and multicellular diploid ($2n$) generations alternately produce each other (see Figure 13.6b). The diploid plant, the *sporophyte*, produces haploid spores by meiosis. These spores divide by mitosis, giving rise to multicellular *gametophytes*, the male and female haploid plants that produce gametes (sperm and eggs). Fertilization, the fusion of gametes, results in a diploid zygote, which divides by mitosis and forms a new sporophyte. In angiosperms, the sporophyte is the dominant generation: It is larger, more conspicuous, and longer-lived

▼ **Figure 38.2** The structure of an idealized flower.



than the gametophyte. The key traits of the angiosperm life cycle can be remembered as the “three Fs”—*f*lowers, double fertilization, and *f*ruits. We'll begin with flowers.

Flower Structure and Function

In angiosperms, the **flower** is the sporophytic structure specialized for sexual reproduction. A flower is typically composed of four types of floral organs: **carpels**, **stamens**, **petals**, and **sepals** (Figure 38.2). When viewed from above, these organs take the form of concentric whorls. Carpels form the first (innermost) whorl, stamens the second, petals the third, and sepals the fourth (outermost) whorl. All are attached to a part of the stem called the **receptacle**. Flowers are determinate shoots; they cease growing after the flower and fruit are formed.

Carpels and stamens are sporophylls—modified leaves specialized for reproduction (see Concept 30.1); sepals and petals are sterile modified leaves. A carpel (megasporophyll) has an **ovary** at its base and a long, slender neck called the **style**. At the top of the style is a sticky structure called the **stigma** that captures pollen. Within the ovary are one or more **ovules**, which become seeds if fertilized; the number of ovules depends on the species. The flower in Figure 38.2 has a single carpel, but many species have multiple carpels. In most species, the carpels are fused, resulting in a compound ovary with two or more chambers, each containing one or more ovules. The term **pistil** is sometimes used to refer to a single carpel or two or more fused carpels (Figure 38.3). A stamen (microsporophyll) consists of a stalk called the filament and a terminal structure called the **anther**; within the anther are chambers called microsporangia (pollen sacs) that produce

▼ **Figure 38.3** The relationship between the terms **carpel** and **pistil**. A simple pistil consists of a single, unfused carpel. A compound pistil consists of two or more fused carpels. Some types of flowers have only a single pistil, while other types have many pistils. In either case, the pistils may be simple or compound.

